

Participation Costs, Social Capital, and the Limits of Mobility Policy

A Model of Status Competition under Income Inequality

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Abstract

Status-related expenditure rises with income inequality, contrary to post-war predictions. We show that this empirical regularity is the generic equilibrium outcome once participation in status competitions is treated as an endogenous, income-constrained decision. In the model, a single high-income participant competes against Q low-income participants through tournaments combining innate ability and acquired social capital, where social capital requires a threshold expenditure. Under diminishing wealth utility, only two stable Nash equilibria exist: universal participation and elite-only participation. A closed-form participation threshold identifies the joint configuration of income differences and participation costs that sustains each equilibrium, and reveals why higher inequality and higher status expenditure can coexist: wider income gaps simultaneously strengthen the high-income participant's investment incentive and price out low-income challengers. For policy, two implications follow. First, reducing participation costs and raising the meritocratic weight on raw talent are complements, not substitutes: neither instrument alone can raise class-level mobility above a hard ceiling of $\mu/[2(1-\mu)]$, where μ is the weight on talent relative to social capital. Second, this ceiling is the binding constraint precisely in the empirically relevant regime — large competitions with many challengers — where both the equilibrium and collective mobility converge to it simultaneously. We characterise the precise conditions under which a third equilibrium, in which only low-income participants invest, emerges and collapses as competition size grows.

Keywords: status competition; income inequality; social capital; participation threshold; mobility ceiling; public policy

JEL classification: D31, J62, Z13

1 Introduction

Whether income inequality intensifies or dampens competition for social position is an old question with a surprisingly durable empirical answer. Post-war scholars — Hirsch (1977), Galbraith (1958), and Frank (1985) — predicted that rising living standards and narrowing income gaps would intensify positional rivalry, as more participants could afford to compete for a fixed stock of social rank. Contemporary evidence consistently finds the opposite: status-related expenditure rises with inequality rather than falling with it, with education expenditure in China [Jin et al., 2011] or data cross multiple countries [Chai et al., 2019, Jaikumar and Sarin, 2015] providing systematic documentation.

This paper concerns itself with the specific form of status expenditure that has direct consequences for economic outcomes: investment in educational credentials and institutional affiliation. Credentials are positional goods in the sense of Hirsch [1977]: their competitive value derives from the scarcity of the positions they unlock, not solely from the human capital they embody. In Bourdieu’s framework [Bourdieu and Passeron, 1977, Bourdieu, 1986], credentials function as institutionalised cultural capital: they confer competitive advantage in selection processes through the social recognition they command rather than solely through the productivity they embody. Rising inequality raises both the stakes of winning a credentialled position and the structural advantage of those who can afford the threshold investment required to compete — which is why education expenditure and inequality may move together in the data. The present model treats education as such a credential rather than as a productive asset that accumulates individual capacity across jobs.

The source of the discrepancy, we argue, lies in a common simplification: existing models of status competition treat the participation decision as a continuous intensity choice — agents decide how much to invest in status-enhancing social capital, with income affecting the level of investment but not whether entry is feasible at all. The threshold cost structure that makes participation a discrete binary decision — affordable or not, depending on income — is seldom explored in this literature. Making participation subject to a threshold budget constraint — the single architectural change in the present paper — breaks the symmetry: wider income gaps simultaneously strengthen the high-income participant’s investment incentive and price out low-income challengers. The positive association between inequality and status expenditure then follows as the generic equilibrium outcome, not a special case.

The model’s specific architecture — the separation of talent μr_i from social capital $(1 - \mu)s_i$ as distinct components of competitive rank — is motivated by several empirical patterns in the elite mobility literature that the standard Becker-Tomes human capital framework doesn’t generate without auxiliary assumptions about non-productive forms of capital. Lo Bello and Morchio [2022] find that dynastic workers in the UK who follow their parent’s occupation find jobs faster but earn *lower* wages than comparably qualified non-dynastic workers; human capital theory predicts wage parity or a premium, not a penalty. Laurison and Friedman [2016] show that the class-origin pay gap in British elite occupations varies sharply by occupation type: it is close to zero in science, academia, and the built environment but approximately 20% in finance, law, and accountancy — the opposite of the human capital prediction that the gap should be largest where

inherited knowledge is most valuable. The data are directly consistent with the model’s prediction that the class gap is increasing in $(1 - \mu)$, the social capital weight: finance and law are low- μ occupations where networks and cultural fit dominate, while science and engineering are high- μ occupations where technical output is directly measurable. Mocetti et al. [2022] exploit two waves of liberalisation of professional services in Italy and find that deregulation reduces intergenerational occupational persistence, with the effect concentrated among less able dynastic workers (proxied by delayed school qualification) — consistent with the model’s prediction that social capital weight $(1 - \mu)$ determines how far raw talent shortfalls can be compensated by positional advantages, so that low- r_i incumbents are the first to exit when regulatory protection shrinks.¹ These residuals share a common structure: each identifies a component of elite access that facilitates entry without raising productivity, and whose weight varies across occupations. That component is the social capital weight $(1 - \mu)$ in the present model.

Against this backdrop, the paper develops the following model. The model considers a status competition in which one high-income participant competes against $Q \geq 1$ low-income participants through a probabilistic tournament combining innate ability and acquired social capital. Social capital — the skills, credentials, and network affiliations that institutional engagement provides — can be obtained only by paying a threshold cost $\bar{v}$, making it accessible to the high-income participant by default and contingent on budget feasibility for low-income participants. The probability of advancement depends on both components, with social capital weighted more heavily than raw talent, following the empirical literature on elite hiring [Rivera, 2015, Bills, 2003]. Status enters utility indirectly through its effect on future income, following Postlewaite [1998]; no separate taste for rank is required.

The model yields four results with direct policy content. First, under diminishing wealth utility only two stable Nash equilibria exist: universal participation (both invest) and elite-only participation (only the high-income participant invests). The transition between them is governed by a participation threshold $\bar{v}^*$ that decreases in both competition size Q and the social capital weight $(1 - \mu)$. Wider inequality sustains high-income investment even as it prices out low-income challengers, which is why status expenditure and inequality rise together.

Second, collective low-income mobility converges to a hard ceiling of $\mu/[2(1 - \mu)]$ in the empirically relevant regime. This ceiling is bounded strictly below one-half for all admissible μ , and falls as social capital is weighted more heavily. It represents a structural limit that no participation-boosting policy can overcome without also raising μ . Reducing participation costs alone expands the range of inequality levels consistent with universal participation, but the ceiling itself is determined by the design of the competition, not the number of participants.

Third, these two instruments are complements in a precise sense: the participation threshold $\bar{v}^*$ is increasing in A , a composite parameter embedding both Q and μ , and the ceiling $\mu/[2(1 - \mu)]$ is increasing in μ . Policies that lower $\bar{v}$ (credential subsidies, open-

¹A related finding concerns UK university admissions, where personal statements — a threshold social capital investment requiring coaching and institutional support — have negligible predictive validity for university performance [Jones, 2013], and students from less advantaged backgrounds out-perform peers from more advantaged backgrounds with the same entry grades. This is consistent with the model’s architecture but pertains specifically to education admissions rather than labour market competition.

access qualifications, reduced network barriers) shift the equilibrium toward universal participation; policies that raise μ (standardised assessment, transparent merit criteria) raise the ceiling at which collective mobility converges. Neither alone is sufficient.

Fourth, a third equilibrium — in which only low-income participants invest while the high-income participant free-rides on raw talent — requires reference-dependent preferences inverting the standard concavity ordering, and is viable only in small competitions ($Q = 1$ or 2). This provides a formal account of the over-investment in elite credentials observed among aspiring candidates from lower-income backgrounds in narrow selection processes [Rivera, 2015]: the mechanism is both preference-dependent and competition-size-specific, and cannot survive scaling to large standardised assessments.

These results position the paper relative to its closest predecessors as follows. The two closest quantitative predecessors each reach conclusions that diverge from the present paper by fixing the participant pool. Frank [1984] shows that wider wage dispersion reduces competition intensity — workers accept wage penalties to occupy the top of the local rank distribution, so narrowing inequality intensifies rivalry. Cole et al. [1992] derive status concern from the marriage market and show directly that agents from a more compact income distribution save more, again implying that narrowing inequality raises status expenditure. In both cases income distribution shapes the intensity of competition within the participant pool, but participation itself is not subject to a threshold budget constraint — all agents compete, and the question is how intensely. Both models therefore predict that *narrowing* inequality raises status expenditure, the opposite of the empirical finding. The present paper introduces a threshold participation cost that makes entry a discrete binary decision, and shows that this single change reverses the sign: *widening* inequality raises status expenditure by sustaining high-income investment while pricing out low-income challengers.

The closest predecessor that the paper differs from is Gallice and Grillo [2019], who consider a model in which educational investment entails productivity gains, signaling power, and social status simultaneously, with status depending on relative achievement in one of three dimensions: innate skills, level of schooling, and income. They characterise conditions under which social concerns increase or decrease educational and income inequality. The present paper diverges from Gallice and Grillo [2019] in two ways. First, the paper uses a mechanism where social capital enters the composite score directly as a productive rank-building input with effects on economic outcome - rather than as a signaling game. Second, the paper uses a threshold participation cost to arrive at binary equilibrium outcomes and implications for mobility ceiling. Whether inequality amplifies or dampens status investment conditional on universal participation is a distinct question from whether inequality sustains or destroys participation itself.

The threshold cost structure used in the paper is motivated by broader literature on social capital and credential investment [Jencks, 1979, Heckman and Mosso, 2014]. Participation costs in elite competitions — tuition fees, membership dues, network cultivation — are threshold rather than continuous expenditures: they must be paid in full to unlock the social capital they confer. This threshold structure is central to the model’s ability to generate binary equilibrium outcomes and a closed-form participation condition. The sociological literature [??Mijs and Savage, 2020] motivates the empirical puzzle and the Case II over-investment result, but the model does not require the sociological apparatus: all results follow from standard expected-utility

maximisation with a threshold entry cost.

The paper proceeds as follows. Section 2 develops the model, deriving winning probabilities for all four investment cases under general Q , characterising the Nash equilibria, deriving the participation threshold in closed form, and establishing the mobility ceilings. Section 3 draws out the policy implications, mapping the model's parameters onto observable competition structures. Section 4 concludes.

2 Model

2.1 Basic Setup

We consider a status competition in which a single high-income participant competes against $Q \geq 1$ low-income participants ($Q + 1$ total). The high-income participant occupies the high-income position y_2 and seeks to maintain it; each low-income participant receives income $y_1 < y_2$ and seeks promotion. Two characteristics determine performance in the competition. First, *raw talent* $r_i \sim \text{Uniform}[0, 1]$, which cannot be affected by expenditure. Second, *social capital* $s_i \sim \text{Uniform}[0, 1]$, which can be enhanced only by paying a threshold cost $\bar{v}$. Both r_i and s_i are independently distributed and unknown to participant i before the competition. The participant with the highest composite score $\mu r_i + (1 - \mu)s_i$ wins the high-income position, where $\mu \in (0, 1)$ weights raw talent relative to social capital.

Since all Q low-income participants share the same income y_1 and all face the same investment decision, we analyze the participation problem from the perspective of a representative low-income participant. The high-income participant's decision is analogous. A participant pays $\bar{v}$ to access social capital (setting $s_i > 0$) or abstains and saves instead (setting $s_i = 0$, relying on raw talent alone). Crucially, no participant knows r_i or s_i prior to deciding whether to pay $\bar{v}$, so investment decisions depend only on the income class of the participant.

Status enters utility only indirectly: the winner obtains disposable income y_2 in the following period, the loser retains y_1 . This treatment, following Postlewaite [1998], preserves a standard wealth-utility framework while endogenising the social dimension of the competition.

The model is designed for a specific class of competitions: those in which a fixed number of high-income positions is contested by a variable pool of challengers, and in which the winner's gain is the loser's foregone advancement. This zero-sum structure is not a simplifying approximation but an accurate description of positional competitions for elite credentials — partnership tracks at professional firms, senior academic appointments, competitive university admissions — where the number of positions is institutionally fixed and occupying one precludes others from doing so. The model does not speak to general labour markets in which education raises productivity across all jobs, nor to settings in which a losing investor retains a human capital gain from the investment. That class of questions — where social capital raises y_1 as well as the probability of winning y_2 — is a natural extension in which the threshold cost generates a productivity externality for non-winners; we leave it to future work. The present model isolates the participation and mobility effects of threshold costs in positional

competitions, which is the mechanism most directly relevant to the empirical puzzle that motivates it.

Focusing on conditions where the contest advantage from social capital investment is substantial, we assume throughout that social capital is weighted more heavily than raw talent:

Assumption 1. $\mu < \frac{1}{2}$

Remark on distributional assumptions. The uniform distributions for r_i and s_i are adopted for tractability: they yield the closed-form trapezoidal CDF $F_X(x)$ whose piecewise structure drives the winning probability derivations in Appendix A. The key qualitative results do not depend on uniformity. The fundamental ordering $p_1 < 1/(Q+1) < p'_1$ — which drives the Nash equilibrium analysis — holds for any pair of distributions that are symmetric on $[0, 1]$ and have the same marginal distribution for r_i and s_i , since the ordering follows from the structural disadvantage of competing without social capital against an opponent who has it, not from the specific shape of the distributions. The closed-form expressions for p_1 , the participation threshold $\bar{v}^*$, and the mobility ceiling $\mu/[2(1-\mu)]$ are specific to the uniform case, but the existence of the binary equilibrium structure (Cases I and IV as the only generically stable equilibria under diminishing utility), and all comparative statics on Q , μ , $\bar{v}$, and $y_2 - y_1$ extend to the broader distributional class. The uniform specification thus functions as a parametric representation of a more general result rather than a special case on which the argument depends.

The distribution of the composite score $x_i \equiv \mu r_i + (1-\mu)s_i$ under Assumption 1 is trapezoidal (derived in Appendix A), with CDF $F_X(x)$ that is piecewise quadratic on $[0, \mu]$, linear on $[\mu, 1-\mu]$, and piecewise quadratic on $[1-\mu, 1]$.

The four investment cases are as follows.

- **Case I** — High-income invests ($v_2 = \bar{v}$), low-income do not ($v_1 = 0$).
- **Case II** — Low-income invest ($v_1 = \bar{v}$), high-income does not ($v_2 = 0$).
- **Case III** — Neither invests ($v_1 = v_2 = 0$).
- **Case IV** — Both invest ($v_1 = v_2 = \bar{v}$).

2.2 Equilibrium Conditions

2.2.1 Winning Probabilities

The derivations are in Appendices A and B. The per-low-income-participant winning probabilities are:

$$\begin{aligned}
(1) \quad p_1 &= \frac{\mu}{2(1-\mu)(Q+2)} && \text{(Case I)} \\
(2) \quad p'_1 &= \frac{6-7\mu}{6(1-\mu)} && \text{(Case II, } Q = 1) \\
(3) \quad p_1 &= \frac{1}{Q+1} && \text{(Cases III and IV)}
\end{aligned}$$

In Case I, the low-income participant competes on raw talent alone against the high-income participant's full composite score — a structural disadvantage that drives p_1 strictly below the symmetric baseline $1/(Q+1)$. In Case II the roles are reversed: the investing low-income participant has the full score advantage, pushing p'_1 strictly above the baseline. For Cases III and IV, symmetry gives $1/(Q+1)$ directly. For general $Q \geq 2$, p'_1 is computable as a rational function of μ of degree Q (Appendix B); all equilibrium results for Case II follow from the ordering result alone, without requiring a closed form for p'_1 .

The high-income participant's probability follows from $p_2 = 1 - Qp_1$ in each case.

The fundamental ordering that drives the Nash equilibrium analysis follows directly from Assumption 1:

$$(4) \quad p_1 < \frac{1}{Q+1} < p'_1 \quad \text{for all } Q \geq 1, \mu < \frac{1}{2}$$

Figure 1 illustrates these probabilities. Panel (a) shows that p_1 (Case I) lies strictly below the baseline $1/(Q+1)$ and decays rapidly with Q . Panel (b) shows p_2 (Case I, high-income participant) lying well above the baseline. Panel (c) shows p'_1 (Case II, $Q = 1$) strictly above the baseline throughout. Panel (d) is discussed in Section 2.2.5.

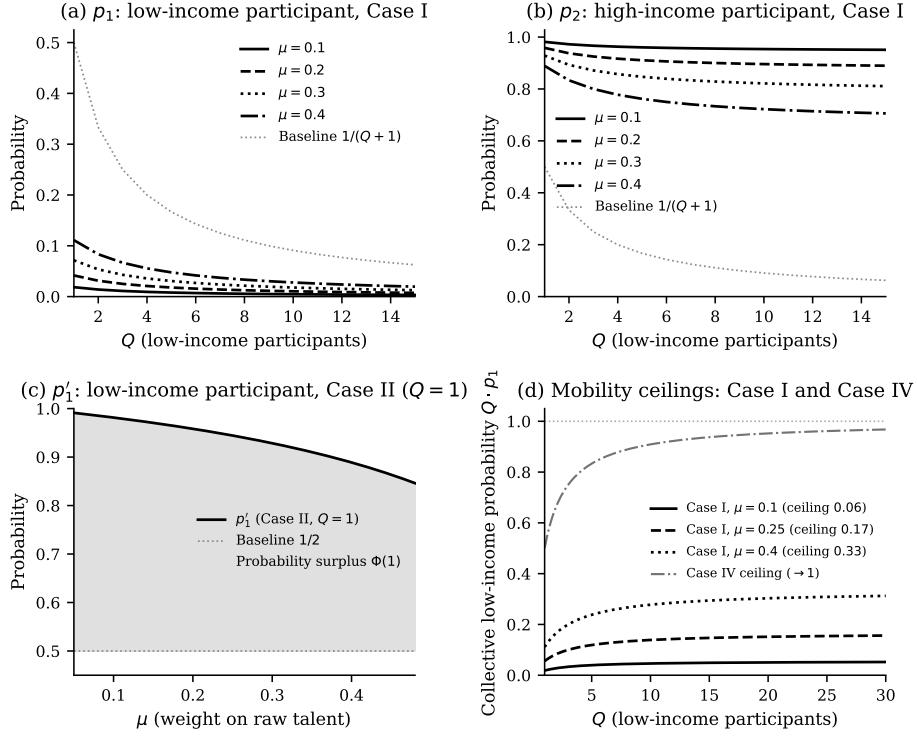


Figure 1: Winning probabilities and mobility ceiling. One high-income participant, Q low-income participants. Dotted horizontal lines in panels (a)–(b) show baselines $1/(Q+1)$ for each Q .

2.2.2 Intertemporal Choice and Utility Payoffs

Participants optimise an intertemporally additive utility $u(\cdot)$ over T periods. In period 1 each participant decides whether to pay $\bar{v}$; from period 2 onward she receives the income corresponding to her position. Define the T -period discount sum $D_T = \sum_{t=0}^{T-2} \delta^{-t} = (1 - \delta^{-(T-1)})/(1 - \delta^{-1})$, where $\delta > 1$ is the discount factor. The case $T = 2$ gives $D_T = 1$ and recovers the standard two-period model. The expected lifetime utilities are:

$$\text{Low-income participant: } u(y_1 - v_1) + \frac{D_T}{\delta} [p_1^c u(y_2) + (1 - p_1^c) u(y_1)]$$

$$\text{High-income participant: } u(y_2 - v_2) + \frac{D_T}{\delta} [p_2^c u(y_2) + (1 - p_2^c) u(y_1)]$$

where p_i^c denotes the winning probability in case $c \in \{I, II, III, IV\}$. Define the utility costs of participation:

$$U_1 \equiv u(y_1) - u(y_1 - \bar{v}), \quad U_2 \equiv u(y_2) - u(y_2 - \bar{v})$$

Under diminishing marginal utility with $y_1 < y_2$: $U_2 < U_1$. Table 1 summarises the four cases as a 2×2 game; full utility payoffs are in Table 4 (Appendix D).

Table 1: Investment strategies

	<i>Low-income withdraws</i>	<i>Low-income invests</i>
<i>High-income withdraws</i>	Case III	Case II
<i>High-income invests</i>	Case I	Case IV

The full utility payoffs for each case are reported in Table 4 in Appendix D; the equilibrium conditions below are derived from those payoffs.

2.2.3 Nash Equilibrium Conditions

For ease of notation define the probability-gap terms:

$$(5) \quad \begin{aligned} p_1 &\equiv \frac{\mu}{2(1-\mu)(Q+2)}, \quad p_2 \equiv 1 - Qp_1 \\ p'_1 &\equiv \frac{6-7\mu}{6(1-\mu)} \quad (Q=1) \end{aligned}$$

The ordering $p_1 < 1/(Q+1) < p_2$ and $p'_1 > 1/(Q+1)$ holds for all $\mu < 1/2$ (Equation 4). Note that $p'_2 = 1 - Qp'_1$ satisfies $p'_2 < 1/(Q+1)$ by the same ordering, but p'_2 does not appear in the equilibrium conditions below: by the algebraic identity $1/(Q+1) - p'_2 = Q(p'_1 - 1/(Q+1))$, it is replaced throughout by $Q \cdot \Phi(Q)$, which is valid for all $Q \geq 1$. We define $\Phi(Q)$ now, before analysing the cases, since it appears in both Case I and Case II:

The *probability surplus* $\Phi(Q)$ is the gain in winning probability a low-income participant obtains from being the sole investor in social capital relative to the symmetric no-investment baseline, scaled by the utility gap and discount factor:

$$(6) \quad \Phi(Q) \equiv \frac{D_T}{\delta} \left(p'_1 - \frac{1}{Q+1} \right) (u(y_2) - u(y_1))$$

Since $p'_1 > 1/(Q+1)$ for all $\mu < 1/2$ (Equation 4), $\Phi(Q) > 0$ always. For $Q = 1$, substituting $p'_1 = (6-7\mu)/[6(1-\mu)]$ yields:

$$(7) \quad \Phi(1) = \frac{D_T}{\delta} \cdot \frac{3-4\mu}{6(1-\mu)} \cdot (u(y_2) - u(y_1))$$

where $(3-4\mu) > 0$ under Assumption 1, so $\Phi(1)$ is strictly positive and decreasing in μ , ranging from $\Phi(1) \rightarrow 1/2$ at $\mu \rightarrow 0$ to $\Phi(1) \rightarrow 1/3$ at $\mu \rightarrow 1/2$. By Proposition 2 below, $\Phi(Q) \sim 1/[Q(Q+1)]$ asymptotically, so the surplus narrows rapidly with competition size.

Case I (high-income invests, low-income withdraws) is a Nash equilibrium if and only if:²

$$(8) \quad \frac{D_T}{\delta} \left(\frac{1}{Q+1} - p_1 \right) (u(y_2) - u(y_1)) < U_1$$

$$(9) \quad U_2 < \frac{D_T}{\delta} \left(p_2 - \frac{1}{Q+1} \right) (u(y_2) - u(y_1))$$

Both conditions hold simultaneously under diminishing utility since $U_2 < U_1$ and $p_2 - 1/(Q+1) \geq 1/(Q+1) - p_1$. A higher $\bar{v}$ favors Case I so long as the high-income participant retains an incentive to invest; a lower $\bar{v}$ can destabilize it if low-income participants are induced to enter. Wider income differences ($y_2 - y_1$ larger) reinforce the high-income participant's incentive to invest while simultaneously raising the stakes enough to attract low-income investment, making the first condition harder to satisfy.

Case II (low-income invest, high-income withdraws). We now state this case as a formal proposition, since it requires conditions beyond standard diminishing utility and its characterisation carries independent theoretical content. The governing object is $\Phi(Q)$, defined above. The probability surplus narrows at rate Q^{-2} : at $Q = 1$ the Case II window is wide ($\Phi(1) \approx 0.46$ at $\mu = 0.2$), but by $Q = 3$ it has collapsed to $\Phi(3) \approx 0.08$ — an 82% reduction — and by $Q = 5$ to $\Phi(5) \approx 0.03$ (Table 3). The qualitative conclusions of Proposition 1 — that Case II requires reference-dependent preferences and that the feasibility window $[U_1, U_2/Q]$ narrows as μ increases — hold for general $Q \geq 1$ by the ordering result in Equation 4.

Proposition 1 (Case II Nash Equilibrium). *Case II (low-income invest, high-income withdraws) is a Nash equilibrium if and only if:*

$$(10) \quad U_1 < \Phi(Q) \leq U_2/Q$$

A necessary condition is therefore $Q \cdot U_1 < U_2$. Under standard diminishing utility with $y_1 < y_2$, $U_2 < U_1$, so $U_2/Q < U_1$ for all $Q \geq 1$ and Condition (10) cannot be satisfied. Case II requires reference-dependent preferences under which $U_1 < U_2$ — the marginal utility cost of $\bar{v}$ is higher for the low-income participant than for the high-income participant, reversing the standard concavity ordering. For $Q = 1$, the condition reduces to:

$$(11) \quad U_1 < \Phi(1) \leq U_2$$

which is feasible if and only if $U_1 < U_2$. For general Q , the necessary condition $Q \cdot U_1 < U_2$ tightens as Q grows; by Proposition 2, $\Phi(Q) \sim 1/[Q(Q+1)]$, so the condition fails for any fixed $U_1 > 0$ beyond $Q^* \approx 1/\sqrt{U_1}$.

Proof. Condition (10) is derived directly from the utility payoffs in Table 4 (Appendix D). The low-income participant prefers to invest given the high-income participant withdraws if and only if $U_1 < \Phi(Q)$. The high-income participant prefers to

²Derived from the payoff table (Appendix D): the low-income participant prefers not to invest, and the high-income participant prefers to invest, given the other's strategy.

withdraw given the low-income participants invest if and only if the gain from deviating to investment, $\frac{D_T}{\delta}(1/(Q+1) - p'_2)(u(y_2) - u(y_1)) = Q \cdot \Phi(Q)$ (using $1/(Q+1) - p'_2 = Q(p'_1 - 1/(Q+1))$), valid for all $Q \geq 1$, is less than the cost U_2 , i.e. $U_2 \geq Q \cdot \Phi(Q)$, equivalently $\Phi(Q) \leq U_2/Q$. Both conditions together require $U_1 < U_2/Q$. Under standard concavity $U_2 < U_1$, so $U_2/Q < U_1$ for all $Q \geq 1$ and the condition fails. $\square \quad \square$

Several features of this proposition deserve comment. First, $\Phi(1)$ is strictly decreasing in μ : as social capital is weighted more heavily, the probability surplus from being the sole investor narrows, and the Case II window must be correspondingly tighter to sustain the equilibrium. At $\mu \rightarrow 0$ (pure raw-talent competition), $\Phi(1) \rightarrow 1/2$, giving the widest possible window; at $\mu \rightarrow 1/2$ (Assumption 1 binding), $\Phi(1) \rightarrow 1/3$. Second, the equilibrium remains fragile even when it exists: higher $\bar{v}$, greater impatience (δ rising), or wider income differences ($y_2 - y_1$ larger) can each collapse it into Case III or Case IV respectively. Third, $\Phi(Q)$ narrows at rate Q^{-2} for general Q (Proposition 2), making Case II quantitatively relevant only for small competitions ($Q = 1$ or 2). This is consistent with the interpretation that Case II — over-investment by aspiring low-income candidates in socially embedded competitions — is a phenomenon of elite selection processes with narrow fields, not of large-scale standardised assessments.

Case III (neither invests) is a Nash equilibrium if:³

$$\begin{aligned} U_1 &> \frac{D_T}{\delta} \left(p'_1 - \frac{1}{Q+1} \right) (u(y_2) - u(y_1)) \\ U_2 &> \frac{D_T}{\delta} \left(p_2 - \frac{1}{Q+1} \right) (u(y_2) - u(y_1)) \end{aligned}$$

Since $p'_1 - 1/(Q+1) \leq p_2 - 1/(Q+1)$ and $U_2 < U_1$, the binding constraint is the second inequality. Case III prevails when participation costs are prohibitively high relative to the income gain. If there is any net utility from competition, Case IV is the likely outcome.

Case IV (both invest) is a Nash equilibrium if:⁴

$$\begin{aligned} (12) \quad U_1 &< \frac{D_T}{\delta} \left(\frac{1}{Q+1} - p_1 \right) (u(y_2) - u(y_1)) \\ (13) \quad U_2 &< Q \cdot \Phi(Q) \end{aligned}$$

With $U_2 < U_1$, this equilibrium rests on the low-income participant's incentive: so long as the low-income participant finds the competition worthwhile, the high-income participant also participates. Note that Equations 8 and 12 are exact complements — Case IV obtains precisely where Case I fails for the low-income participant. The high-income participant's condition $U_2 < Q \cdot \Phi(Q)$ uses the algebraic identity $1/(Q+1) - p'_2 = Q(p'_1 - 1/(Q+1))$, which holds for all $Q \geq 1$ and connects Case IV directly to the

³Both participants prefer withdrawal given the other withdraws.

⁴Both participants prefer investment given the other invests.

probability surplus $\Phi(Q)$ from Proposition 1: the high-income participant's boundary between Cases II and IV is exactly $Q \cdot \Phi(Q)$.

These conditions confirm that under diminishing utility only Cases I and IV are generically stable. For very low $\bar{v}$, both participants invest (Case IV); for very high $\bar{v}$, only the high-income participant invests (Case I); for sufficiently extreme $\bar{v}$, neither invests (Case III). Case II requires a non-standard utility structure and remains fragile throughout.

2.2.4 Participation Threshold

The boundary between Cases IV and I is defined by the critical cost $\bar{v}^*$ at which the low-income participant is indifferent between investing and withdrawing, given the high-income participant invests. Setting Equation 12 as an equality:

$$(14) \quad u(y_1) - u(y_1 - \bar{v}^*) = \frac{D_T}{\delta} \left(\frac{1}{Q+1} - p_1 \right) (u(y_2) - u(y_1))$$

Under log utility $u(y) = \log(y)$, this yields $\bar{v}^* = y_1(1 - (y_1/y_2)^A)$, showing that the threshold depends on the income ratio y_1/y_2 rather than the income level alone; the full derivation is in Appendix C.

Here, the *participation window parameter* A is defined as:

$$(15) \quad A \equiv \frac{D_T}{\delta} \left(\frac{1}{Q+1} - \frac{\mu}{2(1-\mu)(Q+2)} \right) > 0$$

$A > 0$ for all $\mu < 1/2$ and $Q \geq 1$ since $1/(Q+1) > \mu/[2(1-\mu)(Q+2)]$ under Assumption 1. The threshold $\bar{v}^*$ is increasing in A , so any comparative static that reduces A tightens the participation window.

Competition size (Q increasing, more low-income challengers):

$$(16) \quad \frac{\partial A}{\partial Q} = \frac{-\mu(Q+1)^2 + 2(1-\mu)(Q+2)^2}{2(Q+1)^2(Q+2)^2(1-\mu)} \cdot \frac{1}{\delta}$$

Since $\mu < 1$, the denominator $2(Q+1)^2(Q+2)^2(\mu-1) < 0$, while the numerator $N = 2(1-\mu)(Q+2)^2 - \mu(Q+1)^2$ satisfies $N > (Q+2)^2 - \frac{1}{2}(Q+1)^2 = \frac{1}{2}Q^2 + 3Q + \frac{7}{2} > 0$ for all $Q \geq 0$ under $\mu < \frac{1}{2}$. Hence $\partial A / \partial Q < 0$. As the pool of low-income challengers grows, A decreases and $\bar{v}^*$ falls. Equivalently, a larger Q makes Case I increasingly likely to prevail over Case IV. As $Q \rightarrow \infty$, $A \rightarrow 0$ and the participation window closes entirely.

Competition size and real-world competition structures. The comparative statics on Q and μ generate a sharp distinction between small competitions (low Q , low μ) — partner-track interviews, senior academic appointments, narrow graduate schemes — and large competitions (high Q , higher μ) — civil service examinations, mass graduate recruitment, standardised professional qualifications. The empirical interpretation of these comparative statics and their implications for the design of each competition type are developed in Section 3.

2.2.5 Collective Low-Income Probability and the Mobility Ceiling

The analysis so far has focused on the per-low-income-participant winning probability p_1 . A distinct and complementary object is the *collective low-income probability* $Q \cdot p_1$, which measures the probability that any low-income participant displaces the high-income participant — a statement about class-level mobility rather than individual opportunity. Two ceilings emerge depending on which equilibrium prevails.

Case I ceiling. Under Case I, the collective low-income probability in closed form is:

$$(17) \quad Q \cdot p_1 = \frac{Q\mu}{2(1-\mu)(Q+2)}$$

As $Q \rightarrow \infty$:

$$(18) \quad \lim_{Q \rightarrow \infty} Q \cdot p_1 = \frac{\mu}{2(1-\mu)}$$

This ceiling is bounded strictly below $\frac{1}{2}$ for all $\mu < \frac{1}{2}$ and is increasing in μ .

Case IV ceiling. Under Case IV (both invest), all $Q+1$ participants draw from the same distribution F_X , so $p_1 = 1/(Q+1)$ and:

$$(19) \quad \lim_{Q \rightarrow \infty} Q \cdot \frac{1}{Q+1} = 1$$

Under universal participation, collective low-income mobility approaches certainty as the field grows.

Why the Case I ceiling is the economically relevant one. The two ceilings appear to give contradictory messages about the social limits to growth. The resolution lies in the comparative statics: by Equation 16, $\partial A / \partial Q < 0$, so a larger pool of low-income challengers *tightens* the participation window and makes Case I increasingly likely to prevail over Case IV. The Case I ceiling is therefore precisely the one that binds as Q grows: the equilibrium converges toward elite-only participation at exactly the same rate as collective mobility approaches its hard limit, and simultaneously the high-income participant's Case IV incentive condition $U_2 < Q \cdot \Phi(Q)$ weakens as $Q \cdot \Phi(Q) \rightarrow 0$ (Corollary 1). The Case IV ceiling ($\rightarrow 1$) is relevant only when Q is small and costs are low enough to sustain universal participation — the regime where mobility concerns are least acute. It is the Case I ceiling $\mu/[2(1-\mu)]$ that constitutes the binding social limit to growth in the sense of Hirsch [1977].

Simultaneously, the *individual* low-income participant's probability $p_1 \rightarrow 0$ as $Q \rightarrow \infty$: each low-income participant's chances vanish even as collective mobility converges to the ceiling. This divergence between individual and collective mobility — the many trying, the class ceiling binding — is the formal content of the social limits argument.

3 Policy Implications

The model's equilibrium structure carries direct implications for the design of education and credential policy. Three themes emerge: the joint configuration of participation costs and competition design that governs which equilibrium prevails; the mobility ceiling that defines the hard limit on class-level advancement regardless of who participates; and the sharp difference in policy leverage between small and large competitions. We address each in turn.

3.1 Participation Costs and Income Differences as Joint Instruments

The transition between the universal participation equilibrium (Case IV) and the elite-only equilibrium (Case I) is governed jointly by $\bar{v}$ and the income gap $y_2 - y_1$. Two regimes apply.

When $y_1 < \bar{v} < y_2$, the budget constraint is binding: participation is ruled out regardless of expected returns, and the elite-only equilibrium is the only feasible outcome. Only a reduction in $\bar{v}$ large enough to restore $y_1 > \bar{v}$ shifts the equilibrium; marginal cost reductions that leave $y_1 < \bar{v}$ do not change the equilibrium and merely lower the income gap at which the constraint becomes binding in future.

When $\bar{v} < y_1 < y_2$, the constraint is non-binding and the participation threshold $\bar{v}^*$ governs the transition. Under log utility:

$$(20) \quad \bar{v}^* = y_1 \left(1 - \left(\frac{y_1}{y_2} \right)^A \right), \quad A \equiv \frac{D_T}{\delta} \left(\frac{1}{Q+1} - \frac{\mu}{2(1-\mu)(Q+2)} \right)$$

Three policy-relevant comparative statics follow directly from Equation (20).

First, $\bar{v}^*$ depends on the income *ratio* y_1/y_2 rather than the income level alone. A uniform upward shift in both incomes that leaves the ratio constant has no effect on the participation threshold. Policies that raise low-income earnings without compressing the income ratio — for example, minimum wage increases in an economy with proportionally rising top incomes — do not expand the participation window. Policies that directly compress y_2/y_1 , or that lower $\bar{v}$ relative to y_1 , are required.

Second, $\bar{v}^*$ is decreasing in Q (since $\partial A / \partial Q < 0$): a larger pool of low-income challengers reduces each individual's expected return from entry and shrinks the range of costs consistent with universal participation. As competition for a fixed number of elite positions becomes more crowded, the rational response of each low-income participant is more likely to be withdrawal, regardless of the income gap. This comparative static is central to understanding why expanding access to elite competitions — increasing the pool of low-income applicants — can paradoxically reinforce elite-only equilibria unless it is accompanied by a simultaneous reduction in $\bar{v}$.

Third, $\bar{v}^*$ is decreasing in μ : competitions that weight social capital more heavily reduce the expected return to low-income entry, shrinking the participation window. This formalises the intuition behind credentialist barriers: the more decisive institutional affiliation and credentials are relative to raw talent, the narrower the set of cost and income configurations that sustain universal participation.

3.2 The Mobility Ceiling and Its Policy Implications

Even when the universal participation equilibrium is sustained, collective low-income mobility faces a structural ceiling that participation policy cannot remove. Under the elite-only equilibrium (Case I), collective low-income mobility — the probability that any low-income participant displaces the high-income incumbent — converges as $Q \rightarrow \infty$ to:

$$(21) \quad \lim_{Q \rightarrow \infty} Q \cdot p_1 = \frac{\mu}{2(1 - \mu)}$$

This ceiling is bounded strictly below $\frac{1}{2}$ for all admissible $\mu < \frac{1}{2}$, and is increasing in μ . Table 2 reports the ceiling for representative values of μ , alongside the corresponding minimum income ratio y_1/y_2 at which the budget constraint becomes non-binding for a standardised threshold cost.

Table 2: Mobility ceiling and participation window by social capital weight

μ	Ceiling $\mu/[2(1 - \mu)]$	Ceiling (%)	Interpretation
0.10	0.056	5.6%	Social capital near-decisive
0.20	0.125	12.5%	High credential weight
0.30	0.214	21.4%	Moderate credential weight
0.40	0.333	33.3%	Talent and credentials balanced
0.49	0.490	49.0%	Near-maximum under Assumption 1

Ceiling = $\mu/[2(1 - \mu)]$ from Equation (21). Values report the long-run probability that the high-income position is held by a low-income-origin participant, as $Q \rightarrow \infty$ under Case I.

The empirical relevance of these figures is immediate. Intergenerational income mobility studies for the United Kingdom — including Blanden et al. [2005] — find that children born into the bottom income quartile have approximately a 16% probability of reaching the top income quartile as adults. These figures are consistent with the model’s ceiling under $\mu \in [0.20, 0.30]$, a range that corresponds to competitions in which institutional credentials and network affiliations carry substantial but not overwhelming weight relative to raw ability. The model does not claim to calibrate mobility precisely; rather, it identifies the structural parameter μ that would be necessary to account for observed mobility rates, and shows that this parameter is directly amenable to policy intervention through competition design.

The ceiling $\mu/[2(1 - \mu)]$ is the binding constraint in the empirically relevant regime. Under universal participation (Case IV), collective mobility converges to 1 as $Q \rightarrow \infty$ — but this ceiling is only operative when the field is small and participation costs are low enough to sustain universal entry, precisely the conditions least likely to hold in large competitions. It is the Case I ceiling that constitutes the operative social limit to growth as Hirsch (1977) originally argued.

Two policy implications follow directly. The ceiling is determined by μ alone: reducing $\bar{v}$ or expanding the pool of challengers affects which equilibrium prevails

but cannot raise the ceiling itself. Raising μ through standardised, competency-based assessment is therefore a necessary complement to cost reduction. Cost reductions that shift the equilibrium from Case I to Case IV produce a discrete jump in the operative ceiling from $\mu/[2(1 - \mu)]$ to 1; raising μ raises the ceiling continuously within either regime. The largest mobility gains require both instruments applied jointly.

3.3 Small and Large Competitions: Where Policy Leverage Differs

The model's comparative statics on Q and μ generate a sharp distinction between two empirically recognisable types of competition with different policy implications.

Small competitions (low Q , low μ). Partner-track interviews at professional services firms, senior academic appointments, and board-level selections involve a narrow field of challengers and assessors who can observe social capital directly: institutional affiliation, cultural fit, communication style, and network membership are all legible at close range. In the model's terms, these competitions have low Q and low μ .

Two implications follow for this regime. First, the participation window A is relatively wide, and both Cases I and IV are plausible depending on the level of $\bar{v}$. Cost-reduction policies have genuine traction here: since Q is small, the expected return to a low-income participant's investment is non-negligible, and reducing $\bar{v}$ can shift the equilibrium to universal participation. Second, this is the only regime in which the Case II equilibrium — in which low-income participants over-invest while the high-income participant free-rides on raw talent — is empirically relevant. The probability surplus $\Phi(Q)$ is approximately 0.46 at $Q = 1$ and $\mu = 0.2$, giving a genuine feasibility window for the anomalous equilibrium. This is consistent with the documented tendency for aspiring candidates from lower-income backgrounds to over-invest in rank-building credentials — target university choices, internship accumulation, professional society memberships — in precisely the narrow-field selection processes where such credentials are most legible to assessors [Rivera, 2015].

Policy targeting small competitions should therefore combine cost reduction with efforts to lower μ : introducing structured competency-based assessment criteria, reducing the weight placed on network affiliation in hiring decisions, and increasing transparency about what assessors actually value. Without the latter, cost reduction alone may merely extend the over-investment trap to a larger pool of low-income applicants.

Large competitions (high Q , higher μ). National civil service examinations, mass graduate recruitment schemes, and standardised professional qualifications involve wide fields and assessors who must rely on criteria that are verifiable at scale. Standardisation is typically designed to increase μ — to make raw talent more decisive relative to institutional affiliation.

In this regime, two simultaneous forces operate. First, $\partial A / \partial Q < 0$ means the participation window narrows as the field grows: the range of costs sustaining universal participation shrinks regardless of policy intent, because each individual low-income participant's expected return falls with Q . The direction of the equilibrium pressure is always toward elite-only participation as competition size increases.

Second, and working against this, raising μ through standardisation directly raises the ceiling $\mu/[2(1 - \mu)]$. Table 2 shows the magnitude: moving from $\mu = 0.20$ to

$\mu = 0.40$ raises the ceiling from 12.5% to 33.3%. For large competitions where the equilibrium is likely to be Case I regardless of cost levels, this is the operative lever. Access expansion without competition redesign reinforces elite-only equilibria: a larger pool drives A downward without offsetting the structural advantage of the high-income participant. Raising μ through standardisation and anonymisation is the primary instrument in this regime.

3.4 Implications for Education Policy

The model’s primary empirical application is to educational credentials and the competitions they determine entry into. Social capital in the model — the skills, credentials, and network affiliations unlocked by the threshold expenditure $\bar{v}$ — functions as a *productive input into tournament performance*: it directly raises the composite score $\mu r_i + (1 - \mu)s_i$ and thereby directly raises the probability of winning. This is a contest-theoretic mechanism in the tradition of Moldovanu et al. [2007], not a signaling mechanism in the tradition of Spence: there is no asymmetric information about pre-existing types, no separating equilibrium, and no off-path beliefs to be refined. The investment in social capital changes outcomes because it changes performance, not because it reveals anything about the investor to a third-party observer. Two implications for education policy follow from this distinction.

Access policy and rank-building investment. In contest models, expanding access to the productive input raises competitive intensity without necessarily raising class-level mobility. In the present model, subsidising $\bar{v}$ — through means-tested bursaries, state-funded professional qualifications, or open-access credential programmes — expands the set of low-income participants who can afford to invest. If the subsidy is large enough to shift the equilibrium from Case I to Case IV, the ceiling on collective mobility jumps from $\mu/[2(1 - \mu)]$ to the Case IV limit of 1: a discrete and substantial gain. If the subsidy is insufficient to shift the equilibrium — if it reduces $\bar{v}$ without moving it below y_1 — it raises competitive intensity within the Case I regime without improving mobility outcomes. The distinction between margin-shifting and equilibrium-shifting subsidies is therefore crucial for policy evaluation, and it is invisible in models that treat participation as a continuous choice rather than a threshold decision. Because social capital is productive rather than merely informative, the case for credential subsidies rests on their equilibrium-shifting effect rather than on signaling arguments.

Competition design and the weight on social capital. The weight μ is a design parameter of the competition, not a fixed feature of the environment. In educational settings it corresponds to the relative weight placed on raw ability versus acquired credentials, institutional affiliation, and network membership in selection decisions. Standardised examinations, anonymised applications, and competency-based hiring frameworks all correspond formally to an increase in μ : they reduce the contest advantage conferred by the threshold investment and make raw talent more decisive.

The mobility ceiling $\mu/[2(1 - \mu)]$ is directly determined by this design parameter. Because social capital is a productive input and not a signal, the policy instrument for raising the ceiling is competition redesign — changing what the contest rewards — rather than disclosure or information policy. This also resolves the tension between equality of

opportunity and equality of outcome that Heckman and Mosso [2014] identify as central to education policy: the participation threshold $\bar{V}^*$ governs equality of opportunity and responds to $\bar{V}$ (the cost side); the mobility ceiling governs equality of outcome and responds to μ (the design side). These are governed by separate parameters and require separately targeted instruments. Conflating them — treating credential subsidies as if they also raise the ceiling, or treating competition redesign as if it also removes budget constraints — risks improving one dimension without improving the other.

4 Conclusion

This paper develops a formal model of positional competition for elite credentials in the credentialist tradition: social capital is a rank-building input into status tournaments, not a human capital asset that accumulates across jobs. The barrier to entry is budgetary and threshold-structured. Wider income gaps simultaneously sustain high-income investment and price out low-income challengers, generating the empirical pattern — status expenditure rising with inequality — as the generic equilibrium outcome. The post-war prediction that narrowing income gaps would intensify rivalry is not wrong; it corresponds to a different equilibrium, one requiring cost levels low enough for universal participation.

Results. Four principal results emerge. First, under diminishing wealth utility, only two Nash equilibria are generically stable: universal participation (Case IV) and elite-only participation (Case I). The transition is governed by a closed-form participation threshold $\bar{V}^*$ decreasing in both competition size Q and the social capital weight $(1 - \mu)$, with both forces pushing toward elite-only equilibria as competitions grow larger and credentials more decisive.

Second, collective low-income mobility converges to a hard ceiling of $\mu/[2(1 - \mu)]$ in the large- Q regime. This ceiling is bounded strictly below one-half, tightens as social capital is weighted more heavily, and is the operative social limit to growth as Hirsch (1977) originally conjectured. It is now an equilibrium theorem, not a qualitative conjecture.

Third, a reference-dependent equilibrium (Case II) exists in which only low-income participants invest; it is viable only for $Q \leq 2$ and collapses rapidly as competition size grows. This accounts for the documented over-investment in elite credentials by aspiring candidates from lower-income backgrounds in narrow selection processes, a phenomenon specific to small competitions and not replicable in large standardised assessments.

Fourth, competition size and competition design interact in a specific way: as Q grows, the equilibrium shifts toward elite-only participation at exactly the rate collective mobility converges to the ceiling. The two forces are not independent; they arrive together.

Empirical corroboration. Several predictions are already consistent with existing evidence. The Q -comparative static — that access expansion without cost reduction reinforces elite-only equilibria — is corroborated by the Sutton Trust’s finding that the Russell Group share of disadvantaged students declined despite 25 years of widening

participation policy [Sutton Trust, 2023], and by China’s higher education expansion, which left access stratification essentially unchanged [Ding et al., 2021]. The mobility ceiling is consistent with Russell Group access rates of 1–1.4% against a theoretical equality benchmark of 4.4% — a directional and structural correspondence, not a direct numerical mapping. Jerrim and Macmillan [2015] decompose the Great Gatsby Curve using PIAAC data and find that educational inequality accounts for a substantial share of the cross-national mobility gap, motivating access-widening as the primary policy instrument. The present model qualifies this conclusion in a specific way: the participation threshold $\bar{v}^*$ identifies the access channel they recover, but the model shows that reducing $\bar{v}^*$ alone — without simultaneously raising μ — cannot move the mobility ceiling. The three empirical residuals introduced in the introduction — the dynastic wage penalty [Lo Bello and Morchio, 2022], the cross-occupational class pay gap [Laurison and Friedman, 2016], and the Italian liberalisation result [Mocetti et al., 2022] — each map directly onto the model’s μ -($1 - \mu$) architecture, as set out there. In each case human capital theory predicts the wrong sign or cross-sectional pattern; the contest model predicts the correct one.

Policy implications. Two policy instruments emerge and their complementarity is the paper’s central applied finding. Reducing participation costs $\bar{v}$ — through credential subsidies, means-tested bursaries, open-access qualifications, or reduced network barriers — shifts the equilibrium toward universal participation when the reduction is large enough to move the budget constraint from binding to non-binding. This produces a discrete jump in the operative mobility ceiling from $\mu/[2(1 - \mu)]$ to 1. Raising the meritocratic weight μ through standardised assessment, anonymised selection, and competency-based criteria raises the ceiling directly. Cost reduction and competition redesign are complements: the former determines which equilibrium prevails, the latter determines the ceiling that bounds collective mobility under any equilibrium. These instruments are formally independent: the ceiling formula $\mu/[2(1 - \mu)]$ contains only μ , so no reduction in $\bar{v}$, however large, can raise the ceiling without a simultaneous change in competition design. Moreover, sub-threshold reductions in $\bar{v}$ — those insufficient to move the budget constraint from binding to non-binding — raise Q without shifting the equilibrium, which reduces each individual low-income participant’s expected return and can tighten elite-only participation rather than loosen it.

This complementarity implies a specific diagnosis of why two decades of widening participation policy in the United Kingdom — through Office for Students access targets, contextual admissions, and bursary programmes — have produced only modest improvements in elite mobility rates. These policies have primarily addressed $\bar{v}$ at the margin, producing small reductions in financial barriers without the equilibrium-shifting reductions required to move from Case I to Case IV. Simultaneously, the design of elite competition — the weight placed on school type, extracurricular networks, and institutional affiliation in admissions and hiring — has remained largely unchanged, leaving μ and therefore the ceiling intact. The model implies that meaningful improvement in class-level mobility at elite institutions requires policy that addresses both parameters jointly, not sequentially.

The policy prescriptions differ sharply between small and large competitions. In small competitions — partner-track interviews, senior academic appointments, narrow

graduate schemes — both instruments have traction, and the Case II over-investment pattern means that cost-reduction policy alone may extend credential over-investment to a wider pool without raising mobility. The appropriate intervention here combines cost reduction with active reduction of μ : structured competency-based criteria, reduced weight on network affiliation, and transparent assessment rubrics. In large competitions — civil service examinations, mass graduate recruitment, standardised professional qualifications — the participation window closes regardless of cost levels as Q grows. The operative lever is raising μ through standardisation and competition redesign. Expanding the applicant pool without redesigning what the competition rewards reinforces elite-only equilibria in this regime.

Because social capital is a productive rank-building input rather than a signal, the case for credential subsidies rests on their equilibrium-shifting effect, not on signaling arguments. Credential subsidy programmes resisted on the grounds of signaling waste — the argument that expanded credential investment merely inflates the credential floor without generating social returns — are applying the wrong theoretical framework to positional competitions where the investment directly raises competitive rank. In positional contests, the welfare case for subsidies is that they shift the equilibrium to universal participation; it does not depend on whether credentials also signal productivity.

Future empirical work. The model generates testable predictions that go beyond what the existing literature has examined directly. The mobility ceiling $\mu/[2(1-\mu)]$ varies with μ but not with Q : cross-competition variation in the ceiling should be explained by observable proxies for credential weight in selection — the proportion of assessment based on institutional affiliation versus standardised tests, the weight placed on extracurricular social capital in hiring rubrics, the degree to which selection criteria are publicly observable and standardised — but should be orthogonal to competition size once credential weight is controlled for. This is an estimating equation that can be taken to data using existing cross-institutional variation in admission and hiring criteria, and it distinguishes the present model from human capital models in which the ceiling depends on the productivity return to education rather than the contest design parameter.

The central message of the paper is that mobility cannot be expanded by widening access alone. The design of what credential competitions reward and who can afford to enter them are structurally distinct problems governed by distinct parameters — μ and $\bar{v}$ respectively — and the model provides the formal basis for treating them as separate policy objectives requiring separately targeted instruments. Conflating them, as twenty years of widening participation policy largely has, is not merely an oversight; it is a category error that the present framework makes precise.

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A Convolution of Uniform Random Variables

The distribution of a convex combination $w = \mu r + (1 - \mu)s$ of two random variables r, s - which are distributed according to densities $f_R(r)$ and $f_S(s)$ - can be derived using the convolution of two derived variables $x = \mu r$ and $y = (1 - \mu)s$. If r and s are uniformly distributed variables ranging from 0 to 1, then x ranges from 0 to μ and y ranges from 0 to $(1 - \mu)$. The CDF for $F_X(x)$ and $F_Y(y)$ are simply $F_X(x) = \frac{x}{\mu}$, $F_Y(y) = \frac{y}{1-\mu}$ (respectively). Therefore, the densities for x and y are $f_X(x) = \frac{1}{\mu}$ and $f_Y(y) = \frac{1}{1-\mu}$.

The density of w is simple $f(w) = \int_{-\infty}^{\infty} f_Y(w-x)f_X(x)dx$. Since , $f_X(x)$ is $1/\mu$ only between 0 and μ . We have the following integral to evaluate for the convolution

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w-x)dx$$

Since $f_Y(y)$ is 1 only between 0 and $1-\mu$, we have the condition $0 < w-x < 1-\mu \Rightarrow w-(1-\mu) < x < w$. Further, x is distributed only between 0 and μ . As shown in Figure 2, the integral would evaluate to the following⁵ under the assumption $\mu < \frac{1}{2}$.

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w-x)dx = \begin{cases} \frac{1}{\mu} \int_0^w \frac{1}{1-\mu} dx & 0 < w < \mu \\ \frac{1}{\mu} \int_0^{\mu} \frac{1}{1-\mu} dx & \mu < w < 1-\mu \\ \frac{1}{\mu} \int_{w-(1-\mu)}^{\mu} \frac{1}{1-\mu} dx & 1-\mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

This can be simplified as

$$f(w) = \begin{cases} \frac{w}{\mu(1-\mu)} & 0 < w < \mu \\ \frac{1}{1-\mu} & \mu < w < 1-\mu \\ \frac{1-w}{\mu(1-\mu)} & 1-\mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

Figure 2 shows this trapezoidal density with μ set to $\frac{1}{5}$. The CDF $F_X(x)$ can be evaluated by integrating the above density as $P(X < x) = \int_{-\infty}^x f(t)dt$.

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{\mu}{2(1-\mu)} + \frac{x-\mu}{1-\mu} & \mu < x < 1-\mu \\ \frac{\mu^2}{2\mu(1-\mu)} + \frac{2\mu(1-2\mu)}{2\mu(1-\mu)} + \frac{\mu^2-x^2+2x-1}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Simplifying above, we have

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

⁵Notice that as a convex combination, w is distributed only between 0 and 1.

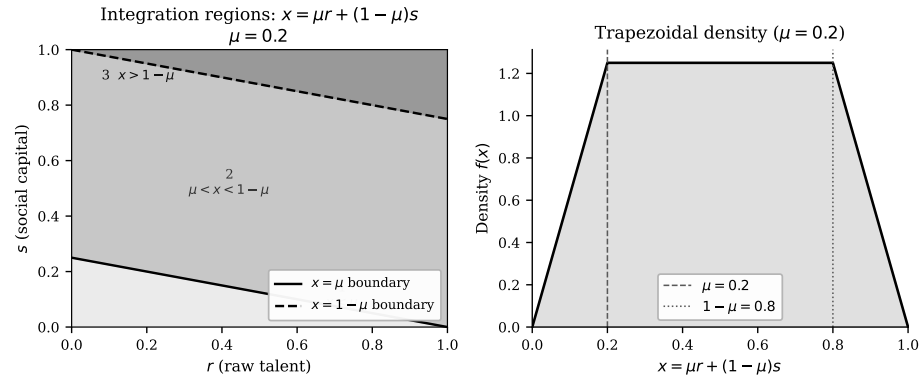


Figure 2: Left: ranges of w and x for convolution of two scaled random variables. Right: trapezoidal density of w as the convex combination with $\mu = \frac{1}{5}$.

B Winning Probability Derivations

Case I: High-income invests, low-income do not

The low-income participant's score is μr_i (no social capital). She wins if μr_i exceeds all $Q - 1$ other low-income participants' scores and the high-income participant's full composite score $x_H = \mu r_H + (1 - \mu)s_H$. Since $\mu r_i < \mu$ for all $r_i \in (0, 1)$ under Assumption 1, only the first piece of F_X applies:

$$P_1(\text{win}|r_i) = r_i^{Q-1} \cdot F_X(\mu r_i) = r_i^{Q-1} \cdot \frac{\mu r_i^2}{2(1-\mu)} = \frac{\mu}{2(1-\mu)} r_i^{Q+1}$$

Integrating over $r_i \sim \text{Uniform}[0, 1]$:

$$p_1 = \int_0^1 \frac{\mu}{2(1-\mu)} r^{Q+1} dr = \frac{\mu}{2(1-\mu)(Q+2)}$$

Case II: Low-income invest, high-income does not

A given low-income participant wins if her full score $x_i = \mu r_i + (1 - \mu)s_i$ exceeds x_k for all $Q - 1$ other low-income participants and exceeds μr_H for the non-investing high-income participant. The conditional probability is:

$$P_1(\text{win}|r_i, s_i) = F_X(x_i)^{Q-1} \cdot G(x_i), \quad G(x) = \begin{cases} x/\mu & 0 < x < \mu \\ 1 & x \geq \mu \end{cases}$$

For $Q = 1$ the double integral over the unit square yields:

$$p'_1 = \frac{6-7\mu}{6(1-\mu)} \quad (Q = 1)$$

For $Q \geq 2$, $F_X(x)^{Q-1} \cdot G(x)$ produces a rational function of μ of degree Q within each CDF region; each specific Q yields a closed form but no compact general formula holds uniformly across all Q .

Cases III and IV

When neither invests, scores are μr_i ; by symmetry each of $Q + 1$ participants wins with probability $1/(Q + 1)$:

$$p_1 = \int_0^1 r^Q dr = \frac{1}{Q+1}$$

When both invest, all scores x_i are identically distributed via F_X ; by symmetry $p_1 = 1/(Q + 1)$ again.

Integration regions

To calculate $\iint P(\text{win}|r,s) dr ds$ over the unit square, we split by the boundaries of the piecewise function. The three regions are shown in Figure 3. If $x = \mu r + (1 - \mu)s$ and $F(x)$ is of the piecewise form with pieces A, B, C :

$$\iint F(x) ds dr = \int_{r=0}^1 \int_0^{\frac{\mu(1-r)}{1-\mu}} A ds dr + \int_{r=0}^1 \int_{\frac{\mu(1-r)}{1-\mu}}^{1-\frac{\mu r}{1-\mu}} B ds dr + \int_{r=0}^1 \int_{1-\frac{\mu r}{1-\mu}}^1 C ds dr$$

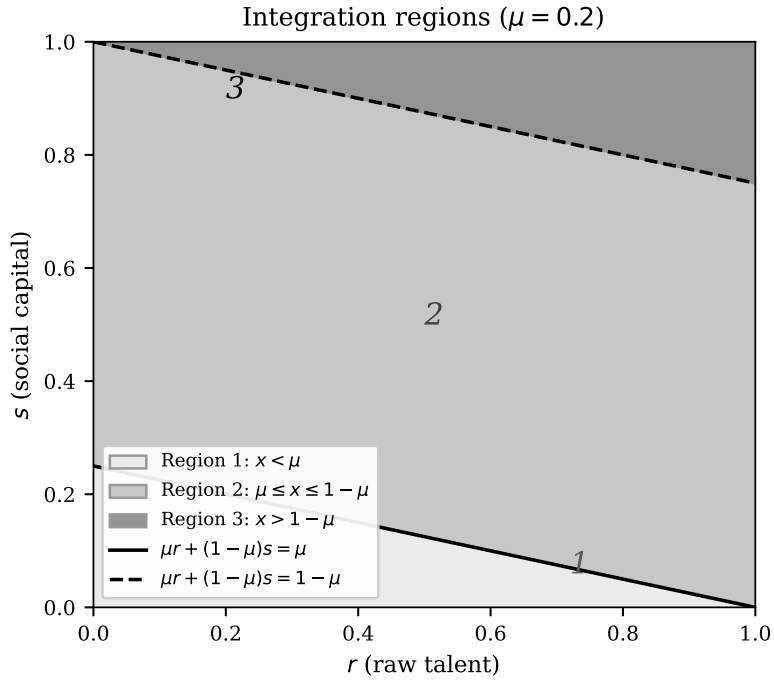


Figure 3: Integration over r, s split by boundaries $\mu r + (1 - \mu)s = \mu$ and $\mu r + (1 - \mu)s = 1 - \mu$

Asymptotic probability surplus

Proposition 2 (Asymptotic probability surplus).

$$\lim_{Q \rightarrow \infty} Q(Q+1)\Phi(Q) = 1$$

equivalently, $\Phi(Q) = \frac{1}{Q(Q+1)} [1 + O(1/Q^2)]$. The convergence is $O(1/Q^3)$ and independent of μ .

Proof. Write $\Phi(Q) = \int_0^1 F_X(t)^{Q-1} [G(t) - F_X(t)] f_X(t) dt$, where the identity follows from $p'_1 = \int F_X^{Q-1} G f_X dt$ and $1/(Q+1) = \int F_X^Q f_X dt$. As $Q \rightarrow \infty$, $F_X(t)^{Q-1}$ concentrates near $t = 1$. Setting $u = 1 - t$ and $c = \mu(1 - \mu)$: $F_X \sim 1 - u^2/(2c)$, $G - F_X \sim u^2/(2c)$, $f_X \sim u/c$. By Laplace's method:

$$\Phi(Q) \sim \frac{1}{2c^2} \int_0^\infty u^3 \exp\left(-\frac{Qu^2}{2c}\right) du = \frac{1}{2c^2} \cdot \frac{2c^2}{Q^2} = \frac{1}{Q^2}$$

Since $c = \mu(1 - \mu)$ cancels, the leading term is μ -independent. The correction is $O(1/Q^3)$. $\square$

Corollary 1. $Q \cdot \Phi(Q) \rightarrow 0$ at rate $1/Q$. The Case II feasibility condition $U_1 < \Phi(Q)$ fails for all $U_1 > 0$ for sufficiently large Q : Case II is asymptotically impossible.

Table 3 reports $\Phi(Q)$ for $Q = 1, \dots, 10$ at representative values of μ , confirming the rapid convergence and μ -independence.

Table 3: Probability surplus $\Phi(Q)$ and asymptotic approximation $1/[Q(Q+1)]$

Q	$\mu = 0.1$	$\mu = 0.2$	$\mu = 0.3$	$\mu = 0.4$	$1/[Q(Q+1)]$	$Q(Q+1)\Phi$
1	0.4815	0.4583	0.4286	0.3889	0.5000	0.917
2	0.1664	0.1651	0.1621	0.1556	0.1667	0.991
3	0.0833	0.0832	0.0829	0.0816	0.0833	0.998
4	0.0500	0.0500	0.0499	0.0497	0.0500	1.000
5	0.0333	0.0333	0.0333	0.0333	0.0333	1.000
6	0.0238	0.0238	0.0238	0.0238	0.0238	1.000
7	0.0179	0.0179	0.0179	0.0179	0.0179	1.000
8	0.0139	0.0139	0.0139	0.0139	0.0139	1.000
9	0.0111	0.0111	0.0111	0.0111	0.0111	1.000
10	0.0091	0.0091	0.0091	0.0091	0.0091	1.000

Values computed via numerical double integration. The last column reports $Q(Q+1)\Phi(Q)$ at $\mu = 0.2$; values at other μ are identical to the precision shown for $Q \geq 4$.

C T-Period Utility Generalisation

The main text presents intertemporal utility under T periods with discount sum $D_T = \sum_{t=0}^{T-2} \delta^{-t}$. We establish here that the qualitative equilibrium structure is unchanged by T .

Closed form for D_T . Since $\delta > 1$, the geometric sum evaluates to:

$$D_T = \frac{1 - \delta^{-(T-1)}}{1 - \delta^{-1}}$$

For $T = 2$: $D_T = 1$, recovering the two-period model. For $T = 3$: $D_T = 1 + 1/\delta$.

Proposition (T-period invariance). *The set of Nash equilibria under T -period utility is identical to that under two-period utility ($T = 2$). The participation threshold $\bar{v}^*$ scales monotonically with D_T : it is strictly increasing in T and strictly decreasing in δ .*

Proof. All Nash equilibrium conditions (Equations 8–13) share the structure:

$$\frac{D_T}{\delta} \cdot \Delta p \cdot (u(y_2) - u(y_1)) \geq U_i$$

where Δp is a probability gap and U_i is a utility cost. Since D_T/δ is a common positive scaling factor, it does not alter which side of any inequality is larger: an equilibrium condition satisfied under $T = 2$ is satisfied for all $T \geq 2$ with the same qualitative direction. The threshold $\bar{v}^*$ (Equation 23) has $A \propto D_T/\delta$; since $(y_1/y_2)^A$ is decreasing in A and $y_1 < y_2$, $\bar{v}^* = y_1(1 - (y_1/y_2)^A)$ increases with A , hence with D_T . D_T is strictly increasing in T (adding positive terms) and strictly decreasing in δ (each term δ^{-t} is decreasing in δ). $\square$

The proposition implies that a longer investment horizon or greater patience raises $\bar{v}^*$, expanding the range of costs under which universal participation (Case IV) is sustained. The two-period model is therefore a conservative baseline: all comparative statics on $\bar{v}^*$ with respect to μ and Q carry over unchanged to the T -period case.

Log utility closed form

Under log utility $u(y) = \log(y)$, the participation threshold $\bar{v}^*$ has a tractable closed form. The indifference condition $U_1 = \frac{D_T}{\delta} \left(\frac{1}{Q+1} - p_1 \right) (u(y_2) - u(y_1))$ becomes:

$$(22) \quad \log \frac{y_1}{y_1 - \bar{v}^*} = A \log \frac{y_2}{y_1}$$

where $A = \frac{D_T}{\delta} \left(\frac{1}{Q+1} - \frac{\mu}{2(1-\mu)(Q+2)} \right) > 0$. Solving:

$$(23) \quad \bar{v}^* = y_1 \left(1 - \left(\frac{y_1}{y_2} \right)^A \right)$$

Since $(y_1/y_2)^A$ is decreasing in A and $y_1 < y_2$, $\bar{v}^*$ is increasing in A , hence decreasing in μ and Q (since $\partial A/\partial \mu < 0$ and $\partial A/\partial Q < 0$). The threshold depends on the income ratio y_1/y_2 rather than the income level alone.

D Utility Payoffs

Table 4 reports the full utility payoffs for each of the four investment cases. In each cell, the upper entry is the high-income participant's utility and the lower entry is the low-income participant's utility. The scale factor D_T/δ multiplies all second-period terms.

Table 4: Utility payoffs (scale factor D_T/δ multiplies all second-period terms)

		<i>Low-income withdraws</i>	<i>Low-income invests</i>
<i>High-income withdraws</i>	High-income	$u(y_2) + \frac{D_T}{\delta} \left[\frac{u(y_2)}{Q+1} + \frac{Qu(y_1)}{Q+1} \right]$	$u(y_2) + \frac{D_T}{\delta} [(1 - Qp'_1)u(y_2) + Qp'_1u(y_1)]$
	Low-income	$u(y_1) + \frac{D_T}{\delta} \left[\frac{u(y_2)}{Q+1} + \frac{Qu(y_1)}{Q+1} \right]$	$u(y_1 - \bar{v}) + \frac{D_T}{\delta} [p'_1u(y_2) + (1 - p'_1)u(y_1)]$
<i>High-income invests</i>	High-income	$u(y_2 - \bar{v}) + \frac{D_T}{\delta} [p_2u(y_2) + (1 - p_2)u(y_1)]$	$u(y_2 - \bar{v}) + \frac{D_T}{\delta} \left[\frac{u(y_2)}{Q+1} + \frac{Qu(y_1)}{Q+1} \right]$
	Low-income	$u(y_1) + \frac{D_T}{\delta} [p_1u(y_2) + (1 - p_1)u(y_1)]$	$u(y_1 - \bar{v}) + \frac{D_T}{\delta} \left[\frac{u(y_2)}{Q+1} + \frac{Qu(y_1)}{Q+1} \right]$

p_1, p_2 : Case I probabilities. p'_1 : Case II low-income probability ($Q = 1$ closed form; general Q numerical). The high-income participant's Case II probability satisfies $1 - Qp'_1$, using the identity $1/(Q+1) - (1 - Qp'_1) = Q(p'_1 - 1/(Q+1))$, valid for all $Q \geq 1$.

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